Performance forecast for Unicharm (TYO: 8113)

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Forecast & Recommendation

After analysis, we predict Unicharm's (TYO: 8113) **Fiscal Year (FY)** Revenue, Operating Income, and Net Profit (all figures in millions of JPY) to be **975,190**, **138,288**, and **94,334** for FY2025. For FY2026, we project figures of **975,190**, **138,204**, and **93,922**, respectively. Our model predicts that financial performance in 2026 will remain effectively flat compared to 2025. Given the lack of projected upside momentum, we recommend a **'Hold'** or **'Do Not Buy'** position.

Methodology: Data Collection

We began by defining a common set of base variables across all analyzed companies (Ezaki Glico, Lion, and Unicharm): Quarters, Revenue, Operating Income, Net Profit, Cost of Goods Sold (COGS), Consumer Price Index (CPI), and Exchange Rate.

In addition to the base variables, we included lagged variables (lags 1 and 4) for Revenue, Operating Income, and Net Profit. These features allow the model to learn from past performance when predicting future results.

We used two types of lags. A **Lag 1** variable represents the previous quarter and helps the model capture momentum. For instance, if Unicharm reported strong diaper and hygiene product sales in Q3, the model learns that this strength often carries into Q4 due to repeat purchases.

A **Lag 4** variable represents the same quarter one year earlier and helps the model capture seasonal patterns. Many consumer goods companies show similar performance at the same time each year, such as stronger sales during year-end periods. For example, Unicharm may experience higher sales in Q4 due to year-end demand and promotional activity.

```python
def create_features(data):
    df_feat = data.copy()

    # Feature 1: Seasonality (Quarter 1-4)
    # Divide Quarters
    df_feat["Quarter"] = df_feat["Date"].dt.quarter

    # Feature 2: Lags (Past Performance)
    # The best predictor of tomorrow is often todat (Lag 1) or this time last year
    for col in ["Revenue", "Operating Income", "Net Profit"]:
        df_feat[f"{col}_Lag1"] = df_feat[col].shift(1) # Previous Quarter
        df_feat[f"{col}_Lag4"] = df_feat[col].shift(4) # Same Quarter Last Year

    # B. MACRO LAGS (Your Request: "Known at time t")
        # We shift these so we predict T using data from T-1
    macros = ['Diapers Exp', 'Cosmetics', 'CPI_YoY', 'Exchange_Rate', "Cardboard", "Sales of daily consumer goods", "COGS"]
    for col in macros:
            df_feat[f'{col}_Lag1'] = df_feat[col].shift(1)

    # Drop the first 4 rows because they now have NaNs (no history for them)
    df_feat = df_feat.dropna().reset_index(drop=True)

    return df_feat

df_model = create_features(df)
```

Fig 1: Feature engineering create_features() function with lag

By including both Lag 1 and Lag 4, the model can account for both recent performance trends and recurring annual demand patterns that are common in consumer goods businesses like Unicharm.

Lagged features that we included ensure that the model only uses historical information available at the time of prediction, thereby preventing look-ahead bias.

Company-Specific Variables

To reflect Unicharm's business structure, we added four external variables aligned with its core segments: Personal Care (Baby, Feminine, Healthcare) and Pet Care.

1. Sales of Cosmetics at Large-Scale Wholesalers

- *Relevance:* Unicharm is part of the broader personal care market, not just basic hygiene. Sales in this category show how confident consumers feel about spending on self-care products. When these sales increase, it suggests consumers are more willing to buy higher-quality or premium products.

2. Diaper Expenditures (Household Spending)

- *Relevance:* Baby diapers and adult incontinence products make up the largest part of Unicharm's sales. This variable directly measures how strong demand is for these products. Changes in this spending reflects shifts in population trends, such as an aging society, as well as Unicharm's ability to raise prices without losing customers.

3. Sales of Daily Household Goods

- *Relevance:* Because Unicharm's products are everyday necessities rather than luxury items, this variable reflects the stable, ongoing demand for its products. It helps the model identify a minimum level of revenue that comes from essential purchases, regardless of broader economic conditions or inflation.

4. Cardboard Collection Dealer Purchase Prices

- *Relevance:* Unicharm relies heavily on packaging and logistics, which use large amounts of cardboard. Changes in cardboard prices give an early signal of whether production costs are likely to increase or decrease.

Strategic Exclusion of Variables: The COGS-Revenue Relationship

During our experimental data analysis, we found a near-perfect correlation between Revenue and Cost of Goods Sold (COGS) (Fig 2). A naive approach might be to include COGS as a predictor to boost the model's accuracy score. However, we purposely excluded it from our revenue forecasting model to maintain logical integrity. The relationship between these variables is one of reverse causation: revenue drives COGS (selling more diapers requires making more diapers), not the other way around. Hence including it would result in data leakage, creating a circular dependency that artificially inflates accuracy while making the model useless for genuine forward-looking prediction.

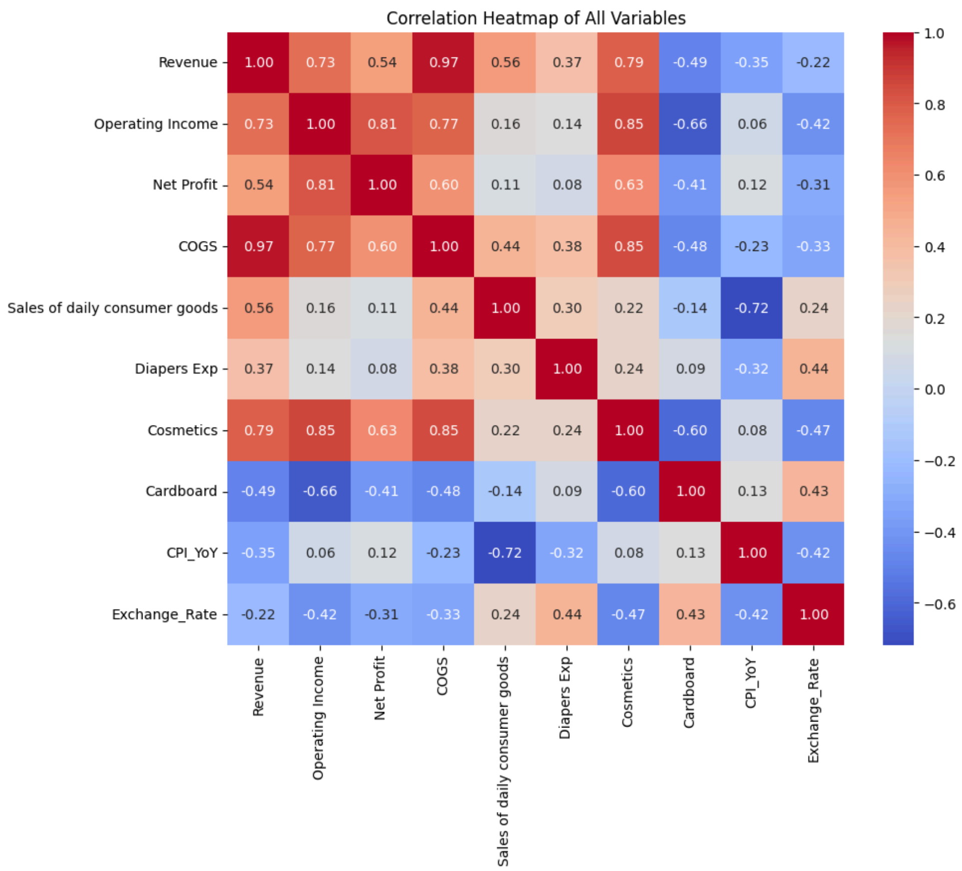

Fig 2: Correlation heatmap of all variables

Model Construction & Strategy

index	Date	Revenue	Operating Income	Net Profit	COGS	Sales of daily consumer goods	Diapers Exp	Cosmetics	Cardboard	CPI_YoY	Exchange_Rate
0	2013/04/01-2013/06/30	143761	17652	10949	78440	8389206	3456	3116	6.5	0	104
1	2013/07/01-2013/09/30	142857	14393	6388	79654	8258224	3468	3075	7.5	-1	100
2	2013/10/01-2013/12/31	152360	17150	12013	83983	9705282	3479	3378	8.5	-1	98
3	2014/04/01-2014/06/30	195709	18517	9080	110729	8618481	3498	2848	9.0	-1	95
4	2014/07/01-2014/09/30	161117	18191	13612	89221	8761662	3516	3059	9.5	0	92

Fig 3: First five rows of the dataset before the create_features() function

The final dataset consisted of 11 unique variables. We handled challenges such as missing data values using linear regression-based imputation. For the forecasting engine, we evaluated two industry-standard machine learning algorithms: Random Forest and XGBoost regression. We chose these methods over classic linear regression because financial data usually contains complex, nonlinear relations between variables, such as how a specific rise in cosmetics demand might disproportionately impact operating income, which simpler models fail to capture. To quantify this, we compared our final Random Forest model against a linear regression model. We found that the linear model's Root Mean Squared Error (RMSE) was, on average, 2.7 times higher than that of the nonlinear model. This massive discrepancy confirmed that a simple linear approach was insufficient for Unicharm's profile.

- **Random Forest** works by building numerous decision trees and averaging their predictions to reduce variance and prevent the model from memorizing noise. Think of this model as a "Consensus Estimate." Instead of depending on a single equity analyst who might be biased (one decision tree), we ask hundreds of independent analysts to make a prediction and then take the average of those. This approach "diversifies" the risk of any single model being wrong and smooths out outliers.
- **XGBoost** builds trees sequentially, where each new tree attempts to correct the errors of the previous ones ("boosting"). Think of this like a Team of experts working to solve a complex problem. The first analyst makes a rough prediction. The second analyst focuses only on where the first one made mistakes and corrects those specific errors. The third fixes the second's remaining errors, and so on. It is a cycle of correcting past mistakes to achieve high precision.

After comparing the two models, Random Forest regression performed better, achieving a lower error. Hence, it was selected to be used for the final projection (as its averaging method demonstrated to be more robust for Unicharm's profile, reducing overfitting and producing more reliable financial forecasts).

To create a consistent multi-year projection free from noise, we employed a rolling forecast loop that iteratively provides predictions forward as inputs. A key component of our strategy was the "Ceteris Paribus" baseline: we assumed all external macroeconomic drivers (such as CPI or Exchange Rates) remained constant at their last known levels. We did this because forecasting macro-indicators introduces significant noise; by holding them constant, we focus on the company's internal performance trends and avoid polluting the model with potentially faulty economic speculations.

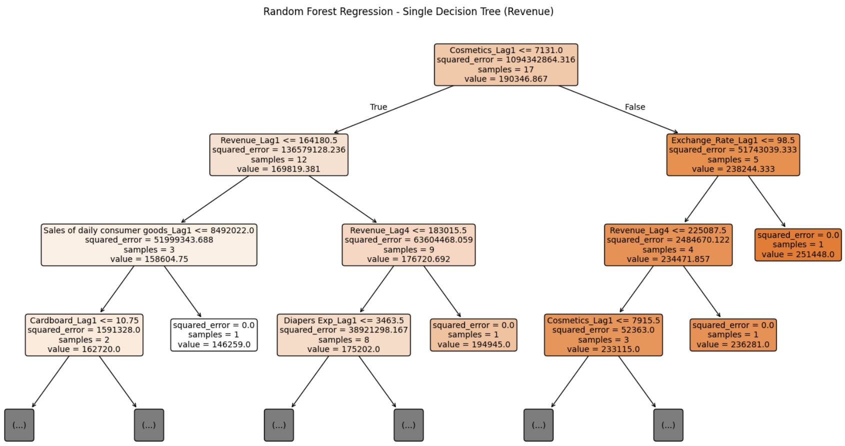

Fig 4: Individual decision tree from the random forest model.

Methodology: Deriving Feature Importance

To understand which inputs were significant for our forecasts, we examined feature importance scores from the Random Forest model. Feature importance measures how much prediction accuracy worsens when a variable's information is disrupted. Variables whose removal causes larger drops in accuracy are considered more important. We used this approach to rank all input variables by their contribution to the model.

Furthermore, we excluded any variable with less than 1% importance as a quality control step. However, since all variables exceeded this threshold, each input provided meaningful information to the forecast.

--- Top Drivers for Revenue ---

Feature	Importance
Revenue_Lag1	0.546527
Cosmetics_Lag1	0.257605
Revenue_Lag4	0.074837
Diapers Exp_Lag1	0.040675
Sales of daily consumer goods_Lag1	0.027322

Fig 5: Feature importance for the revenue forecast

--- Top Drivers for Operating Income ---

Feature	Importance
Cosmetics_Lag1	0.583521
Revenue	0.079379
Operating Income_Lag4	0.069778
Operating Income_Lag1	0.068642
Cardboard_Lag1	0.065273

Fig 6: Feature importance for the revenue forecast

--- Top Drivers for Net Profit ---

Feature	Importance
Operating Income	0.675381
Revenue	0.079226
Sales of daily consumer goods_Lag1	0.067701
Net Profit_Lag1	0.039519
Cosmetics_Lag1	0.034663

Fig 7: Feature importance for the revenue forecast

Feature Importance Analysis

Additional analysis of the model showed different predictors for each financial metric:

- **Revenue:** The most significant predictors were "Revenue_Lag1" and "Sales of Cosmetics". This indicates that recent revenue trends are the most prominent driver of performance, while overall revenue is also affected by conditions in the broader personal care market and consumer spending.
- **Operating Income:** "Sales of Cosmetics" was the strongest predictor, even surpassing past performance (Operating Income_Lag1 and Operating Income_Lag4). This suggests that Unicharm's profitability is heavily reliant on market conditions, with a strong cosmetics market indicating higher consumer demand.
- **Net Profit:** The primary predictors were "Operating Income", "Revenue", and "Sales of Daily Household Goods". While Income and Revenue affirm the link to the bottom line, the significance of "Daily Household Goods" implies that essential commodity demand provides a stable earnings foundation for the company.

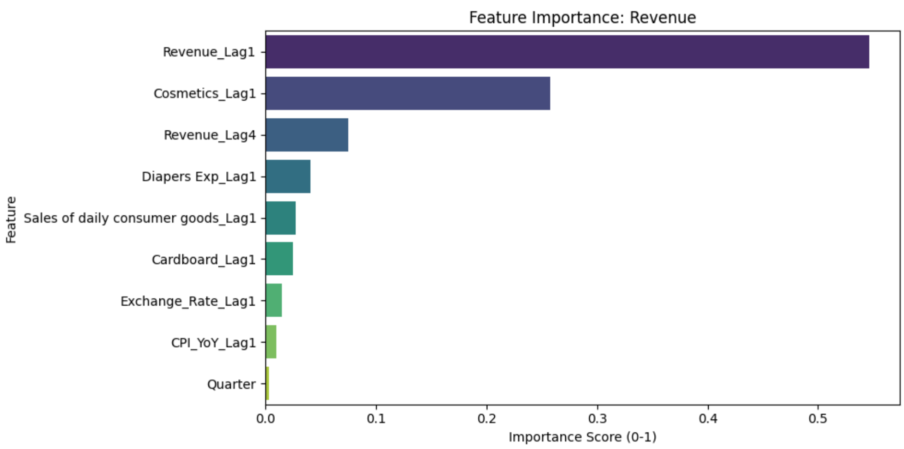

Fig 5: Feature importance for the revenue forecast

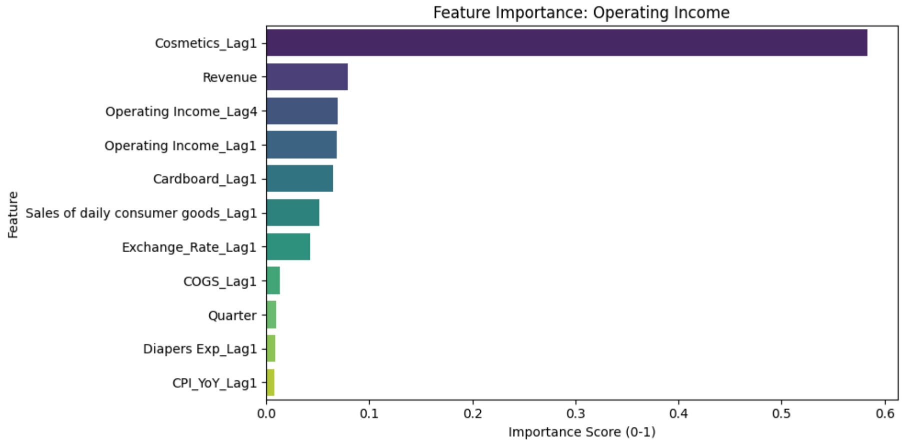

Fig 6: Feature importance for the operating income forecast

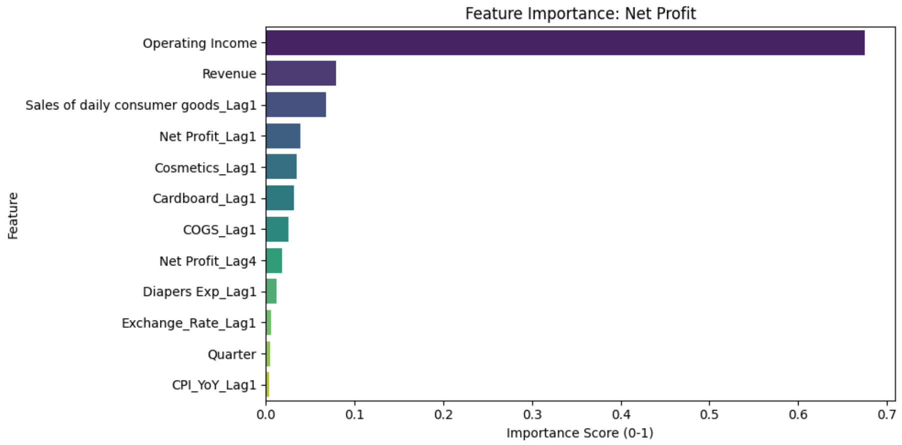

Fig 7: Feature importance for the net profit forecast

Conclusion

Our analysis shows that Unicharm has a stable business built on everyday products that people continue to buy. However, our model does not see strong growth drivers in the near future. The Random Forest forecasts suggest that Unicharm's revenue and profits will stay at roughly the same level through FY2026. Demand for the company's main products is no longer growing quickly, and while cosmetics-related sales help, they are not large enough to drive meaningful overall growth.

We also find that Unicharm's profits are more sensitive to changes in consumer spending than to rising costs. Profits at Unicharm depend more on consumer spending behavior than on input costs. If consumers cut back on discretionary purchases, earnings may fall, while higher raw material costs appear to be a smaller concern.

Until Unicharm shows clear signs of stronger growth such as expanding into new markets, we believe the stock is fairly valued. As a result, we maintain a **Hold** recommendation and advise investors to wait for clearer evidence of long term growth before committing additional capital.

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